CHAPTER 2

Solution of Equations by Determining the Values of the Unknown Function on a Dense Set

2.1. Cauchy's Equations and Jensen's Equation

2.1.1. CAUCHY'S BASIC EQUATION $f(x + y) = f(x) + f(y)$. The equation

$$f(x + y) = f(x) + f(y) \tag{1}$$

was solved by A. L. CAUCHY 1821[12] in essentially the following fashion: From (1), it follows by induction that

$$f(x_1 + x_2 + \cdots + x_n) = f(x_1) + f(x_2) + \cdots + f(x_n), \tag{2}$$

and by putting all $x_k = x$ ($k = 1, 2, \ldots, n$), it follows directly that

$$f(nx) = nf(x). \tag{3}$$

Thus, if $x = (m/n)t$, then,

$$nx = mt$$

and

$$f(nx) = f(mt),$$

and also,

$$nf(x) = mf(t);$$

that is,

$$f\left(\frac{m}{n}t\right) = \frac{m}{n}f(t). \tag{4}$$

[12] A. M. LEGENDRE 1791 and C. F. GAUSS 1809 (cf. 1832[a, b]) used the same reasoning in a less exact form, before A. L. CAUCHY 1821.

If we let $t = 1, f(1) = c$, then,

$$f(x) = cx \tag{5}$$

for every positive rational x.

For $x = 0$, $f(0) = 0$ can be derived immediately from (1); thus (4) and (5) are also valid for $m/n = 0$ and $x = 0$, respectively.

For negative x, we obtain by substituting $y = -x$ in (1)

$$f(x) = f(0) - f(-x) = -f(-x), \tag{6}$$

and thus (4) implies

$$f(rt) = rf(t) \qquad \text{(for all real } t \text{ and all rational } r\text{);} \tag{7}$$

if $t = 1$,

$$f(r) = cr \qquad \text{(for all rational } r\text{).}$$

If $f(x)$ is assumed to be continuous everywhere, then it follows by taking limits on both sides of (5) that

$$f(x) = cx \tag{5}$$

holds for all real x. On the other hand, (5) actually satisfies Eq. (1).

We might remark that *if f is assumed to be continuous and* (1) *satisfied only for positive* (*nonnegative*) *variables, then* (5) *follows for all positive* (*nonnegative*) x.

It is sufficient, as G. Darboux 1875 has shown, to assume *continuity only at a single point* x_0, since it follows from this that $f(x)$ is continuous everywhere. For if

$$\lim_{t\to x_0} f(t) = f(x_0),$$

then for any x, we obtain

$$\begin{aligned}\lim_{u\to x} f(u) &= \lim_{u-x+x_0\to x_0} f[(u - x + x_0) + (x - x_0)] = \lim_{t\to x_0} f[t + (x - x_0)] \\ &= \lim_{t\to x_0} f(t) + f(x - x_0) = f(x_0) + f(x - x_0) = f(x_0 + x - x_0) = f(x),\end{aligned}$$

which was to be proved.

As G. Darboux 1880 has shown also, *it suffices to assume that $f(x)$ is nonnegative* (*nonpositive*) *for sufficiently small positive x*, in order to obtain

(5) as the general solution. Actually, it follows from (1) and from $f(y) \geqslant 0$ for sufficiently small $y > 0$, that

$$f(x+y) = f(x) + f(y) \geqslant f(x),$$

so that $f(x)$ is monotonically increasing (not necessarily strictly monotonically). Furthermore, as we have seen, for rational $x = r$:

$$f(r) = cr$$

[cf. (7).] Now let $\{r_n\}$ be an increasing and $\{R_n\}$ a decreasing sequence of rational numbers converging toward x. Then we have for each n

$$r_n < x < R_n$$

and

$$cr_n = f(r_n) \leqslant f(x) \leqslant f(R_n) = cR_n,$$

from which there immediately follows

$$f(x) = cx, \tag{5}$$

which was to be proved.

On the other hand, let[13] $f(x)$ be bounded *on an arbitrarily small interval* (a, b). On one hand, it follows that

$$g(x) = f(x) - xf(1)$$

also satisfies Eq. (1) and is bounded on (a, b), and, on the other hand, that for any rational r, by (7),

$$g(r) = 0,$$

and hence,

$$g(x+r) = g(x) \qquad (r \text{ arbitrary rational}).$$

However, since it is possible to enter (a, b) from each real x by adding a rational number, $g(x)$ is *bounded everywhere*. If there existed a value x_0, for which

$$g(x_0) = A \neq 0$$

were true, then

$$g(nx_0) = ng(x_0) = nA$$

[13] See also S. Pincherle and U. Amaldi 1901; É. Picard 1928; G. S. Young 1958, among others.

would follow from (3). Hence, for sufficiently large n the function $g(x)$ could assume arbitrarily large values, in contradiction to its boundedness demonstrated above. Thus,

$$g(x) = 0,$$

that is,

$$f(x) = xf(1),$$

Q.E.D. We condense the results of Darboux in

Theorem 1. *If Cauchy's functional equation*

$$f(x + y) = f(x) + f(y) \tag{1}$$

is satisfied for all real x, y, *and if the function* $f(x)$ *is* (a) *continuous at a point*, (b) *nonnegative for small positive* x *-s, or* (c) *bounded in an interval, then*

$$f(x) = cx \tag{5}$$

for all real x. *If* (1) *is assumed only for all positive (nonnegative)* x, y, *then under the same conditions* (5) *holds for all positive (nonnegative)* x.

Later, these conditions were further weakened[14] to the requirement of

[14] H. Hankel 1870; J. L. W. V. Jensen 1878, 1897, 1905, 1906 (boundedness on one side); C. Bourlet 1897; S. Pincherle and U. Amaldi 1901; R. Volpi 1903; G. Peano 1908, 1915[a, b]; R. Schimmack 1908 (integrability); R. D. Carmichael 1911[a, b]; F. Enriques 1913; M. Fréchet 1913, 1914; L. Della Casa 1915; S. Banach 1920; W. Sierpiński 1920[a, b] (measurability); A. A. Fridman 1918, 1949; É. Picard 1922[a, b]; A. Colucci 1926; M. Kormes 1926; H. Stone 1926; R. Caccioppoli 1928; G. Hoheisel 1929[b], G. Hessenberg 1930; H. Reichenbach 1935; M. Kac 1937; C. E. Bonferroni 1938; E. Jacobsthal 1939; B. Kerékjártó 1941[d]; J. Ridder 1941; A. Alexiewicz and W. Orlicz 1945; R. San Juan 1945; E. F. Beckenbach 1948; N. G. de Bruijn 1949, 1952; S. Kurepa 1956[b], 1964[a, c, d]; H. Lenz 1957, 1958[a], 1959; G. S. Young 1958; F. W. Carroll 1959, 1962, 1964; J. M. Koljagin 1959; R. Pyke 1960; H. Rådström 1960; M. Kuczma 1961[a], 1964; D. R. Henney and A. G. Henney 1962; B. Schweizer and A. Sklar 1962[a]; I. Halperin 1963; M. Hosszú 1963[b]; D. R. Henney 1964; W. B. Jurkat 1964, 1965; E. Vincze 1964[a, b, c]; and many others. See also the more comprehensive treatments of J. Thomae 1880; O. Stolz 1886[a]; H. Laurent 1890; C. J. de la Vallée Poussin 1899; S. Pincherle 1906, 1912; P. Dienes 1914; É. Picard 1928; G. H. Hardy, J. E. Littlewood, and G. Pólya 1934; P. Szász 1935; N. Bourbaki 1942, 1947[a, b]; T. Angheluţa 1945[b]; G. M. Fihtengol'c 1949, 1955; V. L. Gončarov 1952[a]; M. Nicolescu 1958; J. Aczél 1960[h], 1961[g], 1962[b]; M. Ghermănescu 1960[b]; H. Lenz 1961; G. Pickert and L. Görke 1962, etc.

boundedness on one side on a set of positive measure,[15] which is equivalent to the *majorizability* (minorizability) of the function $f(x)$ *through a measurable function on a set of positive measure,* while G. HAMEL 1905, with the help of the basis of real numbers constructed for this purpose, succeeded in constructing a solution of (1) that is not of the form (5) and, therefore, is not continuous[16]. (This after an unsuccessful attempt by R. VOLPI 1897.) These results are usually obtained by using rather deep methods from real function theory, so that we cannot reproduce them here. Instead, we shall treat (1) under an assumption of integrability in Section 4.2 and reproduce here only the construction of G. HAMEL 1905.

We make use of the so-called Hamel basis proved by G. HAMEL 1905 with aid of the axiom of choice (cf., for example, W. SIERPIŃSKI 1958) that is of the existence of a nondenumerable set $\boldsymbol{B}$ of real numbers, by which any real number x can be represented uniquely as a finite linear composition

$$x = r_1 b_1 + r_2 b_2 + \cdots + r_n b_n \qquad (b_k \in \boldsymbol{B}) \tag{8}$$

with rational coefficients r_1, r_2, ..., r_n (the coefficients, the choice of basis elements, and the number of terms in the sum depending upon x) and prove:

Theorem 2. *The most general solution which satisfies Cauchy's functional equation*

$$f(x+y) = f(x) + f(y) \tag{1}$$

for all real x, y can be constructed if we choose arbitrary values of f for the points of a Hamel basis $\boldsymbol{B}$ and define $f(x)$ for any real

$$x = r_1 b_1 + r_2 b_2 + \cdots + r_n b_n \qquad (b_k \in \boldsymbol{B}) \tag{8}$$

(the r_k-s are rational) by

$$f(x) = r_1 f(b_1) + r_2 f(b_2) + \cdots + r_n f(b_n). \tag{9}$$

[15] A. OSTROWSKI 1929; H. KESTELMAN 1947, etc.

[16] See also H. BROGGI 1907; H. LEBESGUE 1907, 1950; A. SCHÖNFLIES 1913; A. TERRACINI 1921; W. SIERPIŃSKI 1924, 1958; R. CACCIOPPOLI 1926; E. MACCAFERRI 1939; F. B. JONES 1942[a, b], R. SAN JUAN 1946; G. PICKERT 1949; I. HALPERIN 1951; R. L. WILDER 1952; S. MARCUS 1956[a, b, c], 1957, 1960; H. LENZ 1958[b]; J. ACZÉL AND P. ERDÖS 1965, etc.

Such a solution is continuous if and only if there exists a constant c such that

$$f(b_k) = cb_k$$

for all b_k in the Hamel basis $\boldsymbol{B}$; then, $f(x) = cx$.

The proof is very simple, on the premise of the existence of a Hamel basis: The representation (8) implies by the formulas (2) and (7) (derived without any regularity hypotheses on f) for every real x

$$\begin{aligned} f(x) &= f(r_1 b_1 + r_2 b_2 + \cdots + r_n b_n) \\ &= f(r_1 b_1) + f(r_2 b_2) + \cdots + f(r_n b_n) \\ &= r_1 f(b_1) + r_2 f(b_2) + \cdots + r_n f(b_n), \end{aligned}$$

that is (9). On the other hand, the function (9) constructed by aid of (8) always satisfies functional equation (1).

Finally, the last assertions of Theorem 2 are consequences of Theorem 1, and thus Theorem 2 is completely proved. More abstract algebraic investigations with regard to (1) were carried out by E. NOETHER 1916; N. BOURBAKI 1948; I. N. HERSTEIN 1956, 1957, 1961; G. AUMANN 1957; J. ANDRÉ 1958; O. ZARISKI AND P. SAMUEL 1958; I. N. HERSTEIN AND E. KLEINFELD 1960; K. M. KUTYEV 1963; E. VINCZE 1963[d]; W. S. MARTINDALE 1964, among others.

On the other hand, functional equation (1) can also be investigated for complex functions. Let $f(x) = f_1(x) + if_2(x)$ be a complex function of the real variable x (for example, a function continuous at a point) which satisfies Eq. (1). Then,

$$f_1(x + y) = f_1(x) + f_1(y),$$

$$f_2(x + y) = f_2(x) + f_2(y)$$

hold, and so,

$$f_1(x) = c_1 x, \quad f_2(x) = c_2 x,$$

$$f(x) = (c_1 + ic_2)x = cx;$$

thus, $f(x) = cx$ *with complex c is the most general complex function* (*say, continuous at a point*) *of a real variable, which satisfies functional equation* (1).

We shall return to this same question for complex variables in Sect. 5.1.

2.1.2. THE THREE REMAINING CAUCHY EQUATIONS. The equations[17]

$$f(x+y)=f(x)f(y), \tag{1}$$

$$f(xy)=f(x)+f(y), \tag{2}$$

$$f(xy)=f(x)f(y) \tag{3}$$

(particularly the first) can be solved similarly to 2.1.1(1) in a straightforward manner, but they may also be reduced to 2.1.1(1) very easily.

[17] A. L. CAUCHY 1821; also N. I. LOBAČEVSKIĬ 1829, 1837, 1840, 1855, 1856; J. BOLYAI 1832; W. THOMPSON AND P. C. TAIT 1871; J. L. W. V. JENSEN 1878; E. B. WILSON 1899; R. SCHIMMACK 1908, 1909[c]; A. OSTROWSKI 1917; P. S. MINETTI 1921[a, b]; G. ANDREOLI 1923; O. HAUPT 1928; E. T. BELL 1930; O. HAUPT AND G. AUMANN 1938[a, b]; E. HOPF 1945; C. CHEVALLEY 1946; B. N. DELONE AND D. A. RAĬKOV 1948; S. I. NOVOSELOV 1951; P. M. WOODWARD 1953; S. G. GHURYE 1957; L. L. SEEBECK AND J. W. JEWETT 1957; M. MIKOLÁS 1958, G. POVEDA-RAMOS 1958; L. DABONI 1959; L. S. GURIN 1959; T. W. CHAUNDY AND J. B. MCLEOD 1960; T. ANGHELUŢA 1961[c]; M. KUCZMA 1961[a], 1963[a], 1964; E. F. ASSMUS 1962; J. BERKES 1962; K. KÖNIGSBERGER 1962; E. J. HOPKINS 1963; etc., as well as the comprehensive publications mentioned in footnote 14. If x, y are elements of an algebraic field and the function values are real, then (3) plays a fundamental role in valuation theory (for example, see J. KÜRSCHÁK 1912[a, b], 1913; A. OSTROWSKI 1913, 1917, 1935; W. KRULL 1932; M. DEURING 1935; O. F. G. SCHILLING 1945, 1950; D. ZELINSKY 1948; N. G. ČUDAKOV 1956, and comprehensive treatments by B. L. VAN DER WAERDEN 1930; A. A. ALBERT 1937; G. PICKERT 1951; L. RÉDEI 1954, etc.). If the domain and range of f are both abstract algebraic structures, then (3) is the equation of isomorphisms, homomorphisms, etc. It is clearly not the purpose of this book to discuss these; however see, among others, the last mentioned references. Equations (2) and (3) were investigated for positive integers x, y among others, by M. CIPOLLA 1908, 1915; E. T. BELL 1931[a, b]; P. FRANKLIN 1931[a, b]; A. HERSCHFELD 1931; D. H. LEHMER 1931[a]; I. J. SCHOENBERG 1936, 1962; P. ERDÖS 1938, 1946, 1950, 1954, 1957, 1961; P. ERDÖS AND A. WINTNER 1939; P. ERDÖS AND M. KAC 1940; A. WINTNER 1943, 1944, 1945; N. KABAKER 1946; R. A. RANKIN 1946, 1960, 1962; D. YARDEN AND T. MOTZKIN 1946; R. BELLMAN AND H. N. SHAPIRO 1948; M. KAC 1949; W. J. LE VEQUE 1949; J. MILKMAN 1950[a]; F. PELLEGRINO 1951, 1956; Á. CSÁSZÁR 1952; V. SÓS 1952; A. O. GEL'FOND 1953; L. MOSER AND J. LAMBEK, 1953; H. HALBERSTAM 1955, 1956[a, b]; H. J. KANOLD 1955, 1957, 1960[a, b], 1961, 1962, 1963; I. P. KUBILJUS 1955[a, b, c], 1956, 1959, 1962; S. A. AMITSUR 1956, 1959; J. BALATONI AND A. RÉNYI 1956, 1957; H. DELANGE 1956, 1957[a-d], 1961, 1963; P. T. SHAO 1956; H. N. SHAPIRO 1956; F. SUCCI 1956, 1957, 1960[a, b]; P. COMMENT 1957[a, b, c]; A. M. JAGLOM AND I. M. JAGLOM 1957; A. RÉNYI 1958, 1959[a, b], 1960[b], 1962, 1963; A. RÉNYI AND P. TURÁN 1958; Z. CIESIELSKI 1959; P. TURÁN 1959[a, b], 1963; E. BOMBIARI 1960; R. DEAUX 1960; P. ERDÖS AND A. SCHINZEL 1960; P. ERDÖS AND J. SURÁNYI 1960; E. KISS 1960; V. V. UŽDAVINIS 1960; M. B. BARBAN 1961; E. M. HORADAM 1961; E. WISSING 1961; A. S. BESICOVITCH 1962; K. CORRADI 1962; R. L. DUNCAN 1962; A. A. GIOIA 1962; B. GRIGELIONIS 1962; E. M. PAUL 1962; J. POPKEN 1962; J. ACZÉL AND Z. DARÓCZY 1963[a, b, c]; L. CARLITZ 1963[b]; A. A. MÜLLIN 1963; J. E. SHOCKLEY 1963; T. M. APOSTOL AND H. S. ZUCKERMAN 1964; C. PISOT AND I. J. SCHOENBERG 1964.

For (1), this is done by taking the logarithms of both sides. For this purpose we first observe the validity of the following

Lemma. *If the functional equation* (1) *is satisfied for all real or for all positive x, y, then every solution is either everywhere or nowhere zero.*

For, if there were a y_0 with $f(y_0) = 0$, then it would follow from (1) that

$$f(t) = f[(t - y_0) + y_0] = f(t - y_0)f(y_0) \equiv 0.$$

If (1) is supposed valid only for positive x, y, then this proves $f(t) = 0$ only for $t \geqslant y_0$. But if there were a $t_0 \in (0, y_0)$ such that $f(t_0) \neq 0$, then there would be an n such that $nt_0 \geqslant y_0$; thus, by (1), $0 = f(nt_0) = f(t_0)^n \neq 0$, which is impossible. So, $f(t) \equiv 0$, Q.E.D.

If (1) were supposed just for *nonnegative* x, y, then $f(0) = 1, f(x) = 0$ $(x > 0)$ would still be possible.

We shall disregard these trivial solutions in what follows. On the other hand, we obtain from (1) with $x = y = t/2$

$$f(t) = f\left(\frac{t}{2}\right)^2 > 0. \tag{4}$$

Therefore, any nontrivial solution of (1) is *positive* everywhere. This justifies taking the logarithm on both sides:

$$\ln f(x + y) = \ln f(x) + \ln f(y);$$

this is an equation of the form 2.1.1(1), and thus if f is continuous at a point (or can be majorized by a measurable function on a set of positive measure), then

$$\ln f(x) = cx$$

and

$$f(x) = e^{cx}.$$

Thus we have

Theorem 1.

$$f(x) = e^{cx} \qquad \textit{and} \qquad f(x) = 0 \tag{5}$$

are the most general solutions of

$$f(x + y) = f(x)f(y) \tag{1}$$

(for all real or for all positive x, y), that are continuous at one point (or

that can be majorized by a measurable function on a set of positive measure), while (1) *supposed for nonnegative* x, y *has, in addition to* (5), *the solution*

$$f(0) = 1, \qquad f(x) = 0 \qquad (x > 0)$$

in these classes of functions.

This reasoning, combined with that in Sect. 2.1.1, can also be used for the definition of the exponential function (similarly, that of Theorem 2 below for the definition of the logarithmic function).

It should be remarked that—as (4) shows—minorizability by a measurable function is not sufficient to derive (5); it must be supposed that $f(x)$ be minorized by a *positive* measurable function on a set of positive measure. In fact, all functions of the form

$$f(x) = e^{f_0(x)},$$

where $f_0(x)$ is a discontinuous solution of 2.1.1(1) (the existence of which is assured by 2.1.1, Theorem 2), are solutions of (1) minorized by 0, but they cannot be written in either of the forms (5).

Moreover, the proof of Theorem 1 shows that

$$f(x) \equiv 0 \qquad \text{and} \qquad f(x) = e^{f_0(x)},$$

where $f_0(x)$ *is an arbitrary function satisfying* $f_0(x + y) = f_0(x) + f_0(y)$, *are the most general solutions of* (1).

We shall consider the other two equations,

$$f(xy) = f(x) + f(y) \tag{2}$$

and

$$f(xy) = f(x)f(y), \tag{3}$$

first for *positive* x, y. The substitutions $x = e^u$, $y = e^v$, $f(e^u) = g(u)$ transform (2) and (3) into

$$g(u + v) = g(u) + g(v), \qquad \text{and} \qquad g(u + v) = g(u)g(v),$$

respectively, which were treated before. Thus, *in the above sense, the most general solutions of* (2) *and* (3) *for positive* x, y *are*

$$f(x) = c \ln x \tag{6}$$

for (2) *and*

$$f(x) = e^{c \ln x} = x^c \qquad \text{or} \qquad f(x) = 0 \tag{7}$$

for (3).

If Eq. (2) were satisfied for $x = 0$, it would follow that

$$f(0) = f(x) + f(0)$$

and we would have only the trivial solution

$$f(x) \equiv 0.$$

For $x = 0$, it follows from (3) that

$$f(0) = f(x)f(0),$$

thus, we have in this case either

$$f(x) \equiv 1 \tag{8}$$

or

$$f(0) = 0. \tag{9}$$

If we try to satisfy (2) or (3) for all positive and negative $x \neq 0$, $y \neq 0$, we find immediately, with $x = y = t$ and $x = y = -t$ that

$$2f(t) = f(t^2) = 2f(-t), \qquad \text{hence} \qquad f(-t) = f(t) = c \ln|t|$$

for (2), and

$$f(t)^2 = f(t^2) = f(-t)^2,$$

hence

$$f(-t) = \begin{cases} f(t) = t^c \\ -f(t) = -t^c, \end{cases} \qquad \text{or} \quad 0, \tag{10}$$

for (3), which means that *in this case the most general solutions continuous for* $x \gtrless 0$ *are*

$$f(x) = c \ln|x| \tag{11}$$

for (2), and

$$f(x) = |x|^c, \qquad f(x) = |x|^c \operatorname{sgn} x, \qquad \text{and} \qquad f(x) = 0 \tag{12}$$

for (3); and these, indeed, satisfy the corresponding equations for positive and negative x, $y \neq 0$ — (12) with $c > 0$ also for $x = 0$ or $y = 0$.

If continuity is assumed at only one point, then (11) and (12) remain the general solutions of (2) and of (3) respectively if considered for $x \neq 0$, $y \neq 0$, (because—cf. (10)—if there would be an x and a y such that

$f(-x) = -f(x), f(-y) = f(y)$, then by (3) $f(x)f(y) = f(xy) = f(-x)f(-y) = -f(x)f(y)$); but by (8), (9), and (10), Eq. (3) has all in all the following solutions, continuous at a point, which satisfy Eq. (3) for all real x, y:

$$\left.\begin{array}{lll} f(x) = \begin{cases} |x|^c & (x \neq 0) \\ 0 & (x = 0), \end{cases} & f(x) = \begin{cases} |x|^c \operatorname{sgn} x & (x \neq 0) \\ 0 & (x = 0) \end{cases} & (c \neq 0), \\ f(x) = \quad 0, & f(x) = \quad 1, & \\ f(x) = |\operatorname{sgn} x|, & f(x) = \operatorname{sgn} x. & \end{array}\right\} \tag{13}$$

(It might be remarked that the last two solutions and the first two for $c < 0$, though continuous at a point — in fact at all but one point — are discontinuous at the point 0.)

Thus we have

Theorem 2. *If the functional equation*

$$f(xy) = f(x) + f(y) \tag{2}$$

is valid for all positive x, y, then (6); *if it is valid for all real $x \neq 0$, $y \neq 0$, then* (11) *are its most general solutions that are continuous in a point. If* (2) *were valid for $x = 0$ (or $y = 0$), then*

$$f(x) \equiv 0$$

would be the only solution.

Theorem 3. *If the functional equation*

$$f(xy) = f(x)f(y) \tag{3}$$

is valid for all positive x, y, or for all real x, y, or for all real $x \neq 0$, $y \neq 0$, then (7), (13), *or* (12), *respectively, constitutes the complete set of solutions that are continuous at a point.*

The same results are also true under *measurability* and weaker conditions, and one can also easily express *the most general solutions of (2) and (3) for positive x, y* with aid of the most general solution of 2.1.1 (1): $f(x) = f_0(\ln x)$ *and* $f(x) = e^{f_0(\ln x)}$, 0, *respectively* ($f_0(x + y) = f_0(x) + f_0(y)$).

Here, too, it is possible to generalize to complex-valued functions (cf. Sect. 2.3.5, 3.2.3, 5.1.1). However, for the following theorem the situation is entirely different for real and for complex functions.

Theorem 4.

$$f(x) = x \qquad \textit{and} \qquad f(x) = 0 \tag{14}$$

are the only real functions satisfying both equations

$$f(x + y) = f(x) + f(y) \tag{15}$$

and

$$f(xy) = f(x)f(y) \tag{3}$$

for all positive (nonnegative) x, y, or for all real x, y, or for all real $x \neq 0$, $y \neq 0$.

In fact, (3) implies [cf. (4)] for $x = y = \sqrt{t} \quad (t > 0)$

$$f(t) = f(\sqrt{t})^2 \geqslant 0;$$

that is, f is nonnegative for positive variables and thus, by Theorem 1 of Sect. 2.1.1 as a solution of (15) f can only have the form

$$f(x) = cx.$$

But if we substitute this back into (3) we have

$$cxy = c^2xy,$$

that is,

$$c = 0 \qquad \text{or} \qquad c = 1,$$

and $f(x)$ must have one of the forms (14).

Thus Theorem 4 is proved.

We call attention to the fact that *no regularity assumption* whatever was made in Theorem 4.

On the other hand, it can be proved with aid of the axiom of choice that there exist *noncontinuous complex* solutions of the pair of functional equations (15) and (3) (cf., for example, H. Lebesgue 1907; A. Schönflies 1913; B. Segre 1947; H. Kestelman 1951; W. Sierpiński 1958). The continuous complex solutions of (15) and (3) are

$$f(x) = 0, \qquad f(x) = x, \qquad \text{and} \qquad f(x) = \bar{x},$$

where $\bar{x}$ denotes the compex conjugate of x (cf. Sect. 5.1.1).

2.1.3. Jensen's Equation. The equation[18]

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \tag{1}$$

is the case of equality in Jensen's inequality for convex functions. If the validity of (1) is supposed for all real x, y, then it is easily reduced to 2.1.1(1). For if we put $y = 0$ in (1), we obtain

$$f\left(\frac{x}{2}\right) = \frac{f(x)+f(0)}{2} = \frac{f(x)+a}{2},$$

where $a = f(0)$; thus, with (1) we obtain

$$\frac{f(x)+f(y)}{2} = f\left(\frac{x+y}{2}\right) = \frac{f(x+y)+a}{2}$$

that is,

$$f(x+y) = f(x) + f(y) - a,$$

and with $g(x) = f(x) - a$,

$$g(x+y) = g(x) + g(y),$$

thus, if f is, e.g., continuous at a point,

$$g(x) = cx.$$

Consequently, we have the following

Theorem 1.

$$f(x) = cx + a$$

is the most general solution of

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \tag{1}$$

(supposed to hold for all real x, y) *that is continuous at one point or bounded on one side or integrable or majorizable by a measurable function on a set of positive measure.*

The same argument proves the following

[18] J. L. W. V. Jensen 1905, 1906; H. Reichenbach 1924; D. Pompeiu 1930[b]; J. Ulam 1946; M. Ghermănescu 1958[b]; M. Kuczma 1961[a], 1964; P. A. Moran 1961; H. Jecklin 1962, 1963; J. Aczél and E. Vincze 1963; see also R. P. Boas 1953; P. Erdös and M. Golomb 1955.

Theorem 2. *The most general solution of* (1) *is*

$$f(x) = f_0(x) + a$$

where a is an arbitrary constant and f_0 an arbitrary function satisfying $f_0(x + y) = f_0(x) + f_0(y)$.

2.1.4. Jensen's Equation and Cauchy's Equations for an Interval and for Higher Ranks.[19] *For Jensen's functional equation*

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \tag{1}$$

it is easy to find the general (continuous) solution $f(x) = cx + a$, even if (1) is satisfied *only on an interval,* where we assume f to be continuous for the sake of simplicity. Naturally, this condition can also be weakened. In any case, we make *no assumptions concerning the behavior of f outside of this interval.* It can be assumed that $f(x)$ is defined only on a completely arbitrary interval $[\alpha, \beta]$ (or on a dyadically closed set, which for each pair x, y, also contains $(x + y)/2$)) or, that $f(x)$ satisfies Eq. (1) only there.

In this case, we proceed as follows:

Without any loss of generality we can take $[0, 1]$ as the base interval. We define

$$f(0) = a, \qquad f(1) = b.$$

Then from (1) it follows that

$$f\left(\frac{1}{2}\right) = f\left(\frac{0+1}{2}\right) = \frac{f(0) + f(1)}{2} = \frac{a+b}{2} = a + \frac{1}{2}(b - a),$$

and further,

$$f\left(\frac{1}{4}\right) = \frac{f(0) + f(\frac{1}{2})}{2} = a + \frac{1}{4}(b - a),$$

$$f\left(\frac{3}{4}\right) = \frac{f(\frac{1}{2}) + f(1)}{2} = a + \frac{3}{4}(b - a).$$

[19] See for example C. E. Bonferroni 1938; M. Ghermănescu 1945[a], 1957[b]; M. Hukuhara 1954; J. Aczél and H. Kiesewetter 1957; J. Aczél 1958[a], 1960[e], 1961[f]; J. Aczél, B. Barna, L. Gyarmathi *et al.* 1958; S. Hartman 1958, 1961; P. Erdös 1960; H. Kiesewetter 1961; M. Kuczma 1961[a], 1964; C. Pisot and I. J. Schoenberg 1964.

We now show that, quite generally, for all dyadic x the following holds:

$$f(x) = a + x(b - a) = cx + a$$

$(c = b - a)$. If this has already been proved for dyadic fractions with the denominator 2^n, then

$$f\left(\frac{2m}{2^{n+1}}\right) = f\left(\frac{m}{2^n}\right) = c\frac{m}{2^n} + a = c\frac{2m}{2^{n+1}} + a$$

and

$$f\left(\frac{2m+1}{2^{n+1}}\right) = \frac{f\left(\frac{m}{2^n}\right) + f\left(\frac{m+1}{2^n}\right)}{2} = \frac{c\frac{m}{2^n} + a + c\frac{m+1}{2^n} + a}{2}$$
$$= c\frac{2m+1}{2^{n+1}} + a,$$

which completes the proof. From the continuity assumption, it follows that

$$f(x) = cx + a$$

for all x in the interval $[0, 1]$.

We now assume Cauchy's functional equation

$$f(x + y) = f(x) + f(y) \tag{2}$$

to be satisfied *only for the* x, y *on the interval* $[\alpha, \beta]$, *and* $f(x)$ *to be continuous* on that interval.

In this case, we put

$$x = y = \frac{t}{2} \qquad \left(\frac{t}{2} \in [\alpha, \beta],\ t \in [2\alpha, 2\beta]\right)$$

in (2). Then,

$$f(t) = 2f\left(\frac{t}{2}\right),$$

and with $t = x + y \in [2\alpha, 2\beta]$ (so x and y can move freely in $[\alpha, \beta]$), it follows that

$$f\left(\frac{x+y}{2}\right) = \frac{1}{2}f(x + y) = \frac{f(x) + f(y)}{2},$$

That is, (1) holds in $[\alpha, \beta]$. However, as we have seen, the latter equation has

$$f(x) = cx + a$$

as the only (continuous) solution. If we substitute this in (2), we find that $a = 0$, and have

Theorem 1. *$f(x) = cx + a$ is the general (continuous) solution of*

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \tag{1}$$

and $f(x) = cx$ that of

$$f(x+y) = f(x) + f(y) \tag{2}$$

on the interval $[\alpha, \beta]$, if $[\alpha, \beta] \cap [2\alpha, 2\beta] \neq \emptyset$, else $f(x) = cx + a$.

This problem is related to the one of finding the solutions of the higher rank functional equations

$$f(x_1 + x_2 + \cdots + x_n) = f(x_1) + f(x_2) + \cdots + f(x_n) \tag{3}$$

and

$$f\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right) = \frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n} \tag{4}$$

corresponding to Eqs. (2) and (1).

If functional equation (3) is assumed to be satisfied for all $(x_1, \ldots, x_n)$ or on an interval that contains the origin, we obtain with $x_1 = \ldots = x_n = 0$

$$f(0) = 0,$$

and with $x_3 = \ldots = x_n = 0$,

$$f(x_1 + x_2) = f(x_1) + f(x_2),$$

that is, (2), and this again results in

$$f(x) = cx.$$

The equation treated by C. F. Gauss 1809 originally in connection with the characterization of the normal probability distribution,[20]

$$\begin{aligned} f(x_1) + f(x_2) + \cdots + f(x_n) + f(x_{n+1}) &= 0, \\ x_1 + x_2 + \cdots + x_n + x_{n+1} &= 0, \end{aligned} \tag{5}$$

can be reduced to this. If we put $x_1 = x_2 = \cdots = x_n = 0$, we have $f(0) = 0$, and if we put

$$x_2 = x_3 = \cdots = x_n = 0,$$

we obtain

$$f(x_1) = -f(-x_1);$$

thus,

$$\begin{aligned} f(x_1) + f(x_2) + \cdots + f(x_n) &= -f(-x_1 - x_2 - \cdots - x_n) \\ &= f(x_1 + x_2 + \cdots + x_n), \end{aligned}$$

that is, (3).

However, if (3) or (4) are satisfied only for $x_1, x_2, \ldots, x_n \in [\alpha, \beta]$, which may not contain the origin, then, instead of the above reasoning, we put

$$x_1 = x_2 = \frac{\bar{x}_1 + \bar{x}_2}{2},$$

in order to obtain

$$f(\bar{x}_1 + \bar{x}_2 + x_3 + \cdots + x_n) = 2f\left(\frac{\bar{x}_1 + \bar{x}_2}{2}\right) + f(x_3) + \cdots + f(x_n)$$

and

$$f\left(\frac{\bar{x}_1 + \bar{x}_2 + x_3 + \cdots + x_n}{n}\right) = \frac{2f\left(\frac{\bar{x}_1 + \bar{x}_2}{2}\right) + f(x_3) + \cdots + f(x_n)}{n},$$

respectively.

[20] Cf. also E. Czuber 1891; F. Bernstein 1906, 1907; W. Gosiewski 1909[a, b]; E. A. Borel and R. Deltheil 1923; A. Dias Tavares 1950; Á. Császár 1956[a, b]; M. Hosszú and E. Vincze 1962[a, b], 1963[a, b], etc.

If we compare these equations with (3) and (4) respectively, after omission of the bars, we obtain

$$f\left(\frac{x_1 + x_2}{2}\right) = \frac{f(x_1) + f(x_2)}{2},$$

that is, (1) with the solution

$$f(x) = cx + a,$$

which always satisfies (4), but satisfies (3) only with $a = 0$.

Consequently, we have

Theorem 2. *The general (continuous) solutions of* (3) *and* (4) *on any arbitrary interval are* $f(x) = cx + a$ ($a = 0$ *for* (3) *if* $[\alpha, \beta] \cap [n\alpha, n\beta] \neq \emptyset$).

In Theorem 1, we have assumed the validity of (2) in a *square* $\alpha \leqslant x \leqslant \beta$, $\alpha \leqslant y \leqslant \beta$ (similarly in Theorem 2 that of (3) and (4) in the n-dimensional cube $\alpha \leqslant x_i \leqslant \beta$, $i = 1, 2, \ldots, n$). In Sects. 2.1.1 and 2.1.2, we have also treated the quadrants $x \geqslant 0$, $y \geqslant 0$ as domains. We can also choose other sets for domains of validity; this is a rather new field of investigation concerning functional equations and particularly Cauchy equations (cf. S. Hartman 1958, 1961; P. Erdös 1960; J. Aczél 1961[*f*]; W. B. Jurkat 1964, 1965). Here we treat (2), supposed to hold in the *triangle* $0 \leqslant x \leqslant 1$, $0 \leqslant y \leqslant 1$, $0 \leqslant x + y \leqslant 1$ (1 can be replaced by an arbitrary constant), this time assuming the nonnegativity of $f(x)$ for small positive x, and prove

Theorem 3. *If the functional equation*

$$f(x + y) = f(x) + f(y) \tag{2}$$

is valid for those x, y *which satisfy*

$$0 \leqslant x \leqslant 1, \qquad 0 \leqslant y \leqslant 1, \qquad 0 \leqslant x + y \leqslant 1,$$

and if

$$f(x) \geqslant 0 \tag{6}$$

for small positive x*-values, then*

$$f(x) = cx \tag{7}$$

for all $x \in [0, 1]$.

The proof is a modified version of that for Theorem 1 in Sect. 2.1.1. In fact, 2.1.1(3)

$$f(nx) = nf(x) \tag{8}$$

remains valid for $nx \in [0, 1]$. In particular for $x = 1/n$

$$f(1) = nf(1/n),$$

or with $f(1) = c$,

$$f(1/n) = c/n.$$

(8) again implies

$$f(m/n) = cm/n \qquad \text{for} \quad m \leqslant n,$$

so (7) holds for rational points in the interval (0, 1]. For $y = 0$, (2) gives again

$$f(0) = 0,$$

and the proof of the implications

(6) $\Rightarrow$ monotonicity of f $\Rightarrow$ (7) valid for all real $x \in [0, 1]$
can be taken from the proof of Theorem 1, Sect. 2.1.1. Thus Theorem 3 is proved.

2.2. Equations of the Type $f(x+y) = F[f(x), f(y)]$, Related Equations, and Generalizations

2.2.1. Method of Solution for Equations of the Type $f(x + y) = F[f(x), f(y)]$. The method used with the equation

$$f(x + y) = f(x) + f(y)$$

and with the other Cauchy equations can also be applied to other similar equations. For example,

$$f(x + y) = f(x) + f(y) + f(x)f(y). \tag{1}$$

From this, it follows that

$$f(2x) = 2f(x) + f(x)^2 = [1 + f(x)]^2 - 1, \tag{2}$$

$$f(3x) = f(2x + x) = 3f(x) + 3f(x)^2 + f(x)^3 = [1 + f(x)]^3 - 1,$$